

The *Prisoner's Dilemma* is an example of a game with a strictly dominant-strategy equilibrium. For a detailed account of the history of this game and an in-depth analysis of it see <http://plato.stanford.edu/entries/prisoner-dilemma> or http://en.wikipedia.org/wiki/Prisoner's_dilemma.

An instance of the Prisoner's Dilemma is the following situation. Doug and Ed work for the same company and the annual party is approaching. They know that they are the only two candidates for the best-worker-of-the-year prize and at the moment they are tied; however, only one person can be awarded the prize and thus, unless one of them manages to outperform the other, nobody will receive the prize. Each chooses between exerting *Normal* effort or *Extra* effort (that is, work overtime) before the party. The corresponding game-frame is shown in Figure 2.8.

		Player 2 (Ed)	
		Normal effort	Extra effort
Player 1 (Doug)	Normal effort	o_1	o_2
	Extra effort	o_3	o_4

o_1 : nobody gets the prize and nobody sacrifices family time
 o_2 : Ed gets the prize and sacrifices family time, Doug does not
 o_3 : Doug gets the prize and sacrifices family time, Ed does not
 o_4 : nobody gets the prize and both sacrifice family time

Figure 2.8: The Prisoner's Dilemma game-frame

Suppose that both Doug and Ed are willing to sacrifice family time to get the prize, but otherwise value family time; furthermore, they are envious of each other, in the sense that they prefer nobody getting the prize to the other person's getting the prize (even at the personal cost of sacrificing family time). That is, their rankings are as follows: $o_3 \succ_{\text{Doug}} o_1 \succ_{\text{Doug}} o_4 \succ_{\text{Doug}} o_2$ and $o_2 \succ_{\text{Ed}} o_1 \succ_{\text{Ed}} o_4 \succ_{\text{Ed}} o_3$. Using utility functions with values from the set $\{0, 1, 2, 3\}$ we can represent the game in reduced form as shown in Figure 2.9. In this game exerting extra effort is a strictly dominant strategy for every player; thus *(Extra effort, Extra effort)* is a strictly dominant-strategy equilibrium.

Definition 2.2.4 Given an ordinal game in strategic form, let o and o' be two outcomes. We say that o is *strictly Pareto superior* to o' if every player prefers o to o' (that is, if $o \succ_i o'$, for every Player i). We say that o is *weakly Pareto superior* to o' if every player considers o to be at least as good as o' and at least one player prefers o to o' (that is, if $o \succeq_i o'$, for every Player i and there is a Player j such that $o \succ_j o'$). In reduced-form games, this definition can be extended to strategy profiles as follows. If s and s' are two strategy profiles, then s is *strictly Pareto superior* to s' if $\pi_i(s) > \pi_i(s')$ for every Player i and s is *weakly Pareto superior* to s' if $\pi_i(s) \geq \pi_i(s')$ for every Player i and, furthermore, there is a Player j such that $\pi_j(s) > \pi_j(s')$.

		Player 2 (Ed)	
		Normal effort	Extra effort
Player 1 (Doug)	Normal effort	2 2	0 3
	Extra effort	3 0	1 1

Figure 2.9: The Prisoner's Dilemma game

For example, in the Prisoner's Dilemma game of Figure 2.9, outcome o_1 is strictly Pareto superior to o_4 or, in terms of strategy profiles, *(Normal effort, Normal effort)* is strictly Pareto superior to *(Extra effort, Extra effort)*.

When a player has a strictly dominant strategy, it would be irrational for that player to choose any other strategy, since she would be guaranteed a lower payoff in every possible situation (that is, no matter what the other players do). Thus in the Prisoner's Dilemma individual rationality leads to *(Extra effort, Extra effort)* despite the fact that both players would be better off if they both chose *Normal effort*. It is obvious that if the players could reach a *binding* agreement to exert normal effort then they would do so; however, the underlying assumption in non-cooperative game theory is that such agreements are not possible (e.g. because of lack of communication or because such agreements are illegal or cannot be enforced in a court of law, etc.). Any non-binding agreement to choose *Normal effort* would not be viable: if one player expects the other player to stick to the agreement, then he will gain by cheating and choosing *Extra effort* (on the other hand, if a player does not believe that the other player will honor the agreement then he will gain by deviating from the agreement herself). The Prisoner's Dilemma game is often used to illustrate a conflict between individual rationality and collective rationality: *(Extra effort, Extra effort)* is the individually rational outcome while *(Normal effort, Normal effort)* would be the collectively rational one.